

Tidying Data

tidyr

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tidyr

Functions for tidying data.

What is tidy data?

“Tidy datasets are all alike, but every messy dataset is messy in its own way.” — Hadley Wickham



Tidy Data

country	year	cases	population
Afghanistan	1999	7745	19987071
Afghanistan	2000	8666	20593360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	1272915272
China	2000	216766	1280425583

variables

country	year	cases	population
Afghanistan	1999	7745	19987071
Afghanistan	2000	8666	20593360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	1272915272
China	2000	216766	1280425583

observations

country	year	cases	population
Afghanistan	1999	7745	19987071
Afghanistan	2000	8666	20593360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	1272915272
China	2000	216766	1280425583

values

Each column is a single *variable*

Each row is a single *observation*

Each cell is a *value*

pivot_longer()

```
1 pivot_longer(<DATA>, <NAMES TO>, <VALUES TO>, <VARIABLES>)
```

Lord of the Rings

```
1 lotr <- tribble(  
2           ~film,      ~race, ~female, ~male,  
3   "The Fellowship Of The Ring",  "Elf",  1229L,  971L,  
4   "The Fellowship Of The Ring", "Hobbit",   14L, 3644L,  
5   "The Fellowship Of The Ring",  "Man",    0L, 1995L,  
6           "The Two Towers",  "Elf",  331L,  513L,  
7           "The Two Towers", "Hobbit",   0L, 2463L,  
8           "The Two Towers",  "Man",  401L, 3589L,  
9   "The Return Of The King",  "Elf",  183L,  510L,  
10  "The Return Of The King", "Hobbit",   2L, 2673L,  
11  "The Return Of The King",  "Man",  268L, 2459L  
12 )
```

Lord of the Rings

```
1 lotr
```

```
# A tibble: 9 × 4
```

	film	race	female	male
	<chr>	<chr>	<int>	<int>
1	The Fellowship Of The Ring	Elf	1229	971
2	The Fellowship Of The Ring	Hobbit	14	3644
3	The Fellowship Of The Ring	Man	0	1995
4	The Two Towers	Elf	331	513
5	The Two Towers	Hobbit	0	2463
6	The Two Towers	Man	401	3589
7	The Return Of The King	Elf	183	510
8	The Return Of The King	Hobbit	2	2673
9	The Return Of The King	Man	268	2459



new data alert!



lotr

	film	race	female	male
1	The Fellowship Of The Ring	Elf	1229	971
2	The Fellowship Of The Ring	Hobbit	14	3644
3	The Fellowship Of The Ring	Man	0	1995
4	The Two Towers	Elf	331	513
5	The Two Towers	Hobbit	0	2463
6	The Two Towers	Man	401	3589
7	The Return Of The King	Elf	183	510
8	The Return Of The King	Hobbit	2	2673
9	The Return Of The King	Man	268	2459

Where does it come from?

exercises

source:

github.com/jennybc/lotr-tidyr

How can I use it?

Run the code at the top of
exercises

`View(lotr)`



*this saves it in your
global environment*

pivot_longer()

```
1 lotr |>
2   pivot_longer(
3     names_to = "sex",
4     values_to = "words",
5     cols = female:male
6   )
```


pivot_longer()

```
# A tibble: 18 × 4
```

	film	race	sex	words
	<chr>	<chr>	<chr>	<int>
1	The Fellowship Of The Ring	Elf	female	1229
2	The Fellowship Of The Ring	Elf	male	971
3	The Fellowship Of The Ring	Hobbit	female	14
4	The Fellowship Of The Ring	Hobbit	male	3644
5	The Fellowship Of The Ring	Man	female	0
6	The Fellowship Of The Ring	Man	male	1995
7	The Two Towers	Elf	female	331
8	The Two Towers	Elf	male	513
9	The Two Towers	Hobbit	female	0
10	The Two Towers	Hobbit	male	2463
11	The Two Towers	Man	female	101



new data alert!



table2, table4a, who

	country	iso2	iso3	year	new_sp_m014
1	Afghanistan	AF	AFG	1980	NA
2	Afghanistan	AF	AFG	1981	NA
3	Afghanistan	AF	AFG	1982	NA
4	Afghanistan	AF	AFG	1983	NA
5	Afghanistan	AF	AFG	1984	NA
6	Afghanistan	AF	AFG	1985	NA
7	Afghanistan	AF	AFG	1986	NA
8	Afghanistan	AF	AFG	1987	NA

	country	1999	2000
1	Afghanistan	745	2666
2	Brazil	37737	80488
3	China	212258	213766

	country	year	type	count
1	Afghanistan	1999	cases	745
2	Afghanistan	1999	population	19987071
3	Afghanistan	2000	cases	2666
4	Afghanistan	2000	population	20595360
5	Brazil	1999	cases	37737
6	Brazil	1999	population	172006362
7	Brazil	2000	cases	80488
8	Brazil	2000	population	174504898
9	China	1999	cases	212258
10	China	1999	population	1272915272
11	China	2000	cases	213766
12	China	2000	population	1280428583

Where does it come from?

The `tidyr` R package

How can I use it?

```
library(tidyr)
View(table2)
View(table4a)
View(who)
```



*they're
invisible!*

Your Turn 1

Use `pivot_longer()` to reorganize `table4a` into three columns: `country`, `year`, and `cases`.

Your Turn 1

```
1 table4a |>
2   pivot_longer(
3     names_to = "year",
4     values_to = "cases",
5     cols = -country
6   )
```

A tibble: 6 × 3

	country	year	cases
	<chr>	<chr>	<dbl>
1	Afghanistan	1999	745
2	Afghanistan	2000	2666
3	Brazil	1999	37737
4	Brazil	2000	80488
5	China	1999	212258
6	China	2000	213766

pivot_wider()

```
1 pivot_wider(<DATA>, <NAMES FROM>, <VALUES FROM>)
```

wide

id	x	y	z
1	a	c	e
2	b	d	f

Animation by Mara Averick

pivot_wider()

```
1 lotr |>
2   pivot_longer(
3     names_to = "sex",
4     values_to = "words",
5     cols = female:male
6   ) |>
7   pivot_wider(
8     names_from = race,
9     values_from = words
10  )
```

A tibble: 6 × 5

	film	sex	Elf	Hobbit	Man
	<chr>	<chr>	<int>	<int>	<int>
1	The Fellowship Of The Ring	female	1229	14	0
2	The Fellowship Of The Ring	male	971	3644	1995
3	The Two Towers	female	331	0	401
4	The Two Towers	male	513	2463	3589
5	The Return Of The King	female	183	2	268
6	The Return Of The King	male	510	2673	2459

Your Turn 2

Use `pivot_wider()` to reorganize `table2` into four columns: `country`, `year`, `cases`, and `population`.

Create a new variable called `prevalence` that divides `cases` by `population` multiplied by 100000.

Pass the data frame to a `ggplot`. Make a scatter plot with `year` on the x axis and `prevalence` on the y axis. Set the color aesthetic (`aes()`) to `country`. Use `size = 2` for the points. Add a line geom.

```
1 table2
```

Your Turn 2

```
1 table2 |>
2   pivot_wider(
3     names_from = type,
4     values_from = count
5   ) |>
6   mutate(prevalence = (cases / population) * 100000)
```

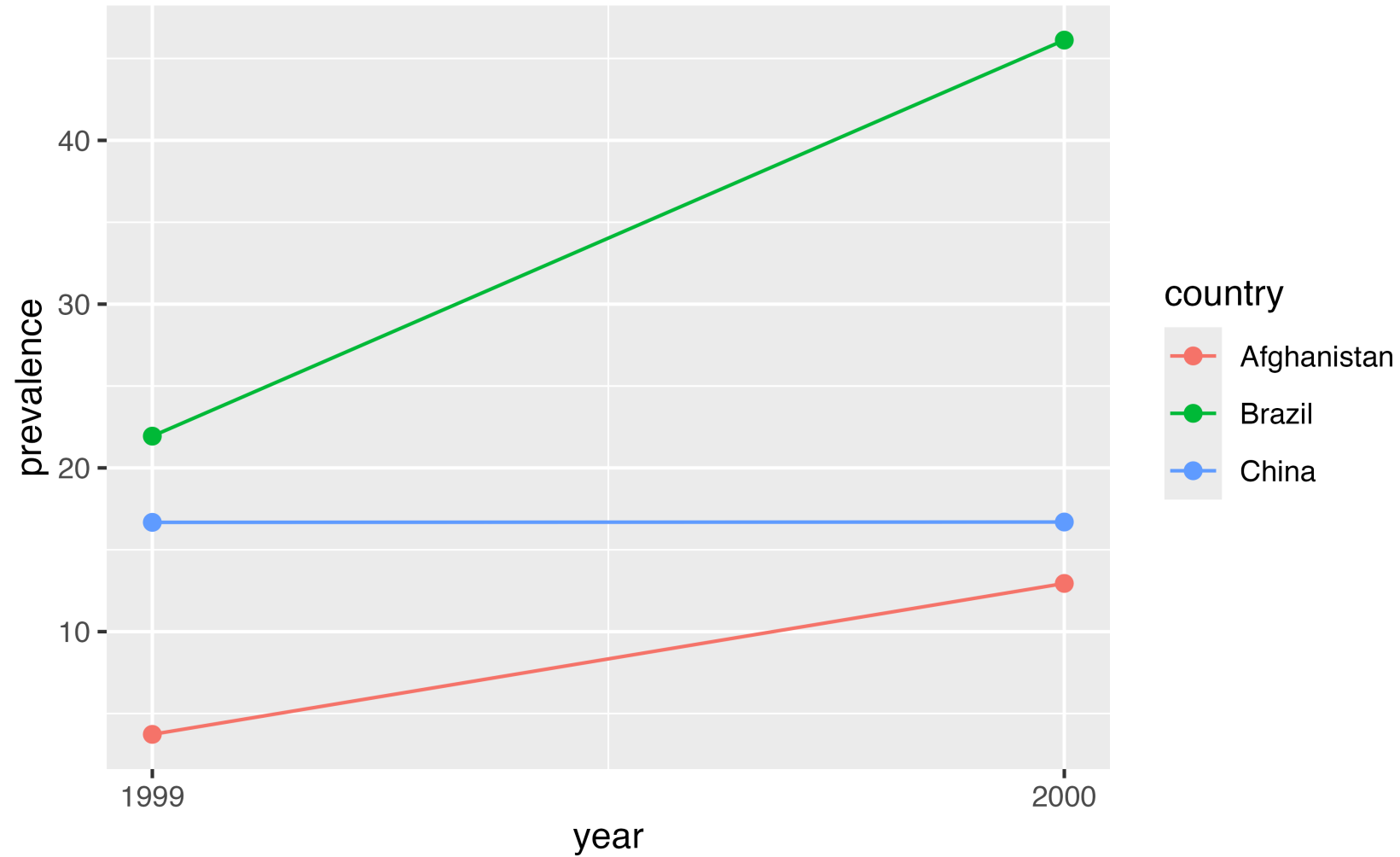
A tibble: 6 × 5

	country	year	cases	population	prevalence
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Afghanistan	1999	745	19987071	3.73
2	Afghanistan	2000	2666	20595360	12.9
3	Brazil	1999	37737	172006362	21.9
4	Brazil	2000	80488	174504898	46.1
5	China	1999	212258	1272915272	16.7
6	China	2000	213766	1280428583	16.7

Your Turn 2

```
1 table2 |>
2   pivot_wider(
3     names_from = type,
4     values_from = count
5   ) |>
6   mutate(prevalence = (cases / population) * 100000) |>
7   ggplot(aes(x = year, y = prevalence, color = country)) +
8   geom_point(size = 2) +
9   geom_line() +
10  scale_x_continuous(breaks = c(1999L, 2000L))
```

Your Turn 2



Your Turn 3

Pivot the 5th through 60th columns of `who` so that the names of the columns go into a new variable called `codes` and the values go into a new variable called `n`. Then select just the `country`, `year`, `codes` and `n` variables.

```
1 who
```

Your Turn 3

```
1 who |>
2   pivot_longer(
3     names_to = "codes",
4     values_to = "n",
5     cols = 5:60
6   ) |>
7   select(country, year, codes, n)
```

Your Turn 3

```
# A tibble: 405,440 × 4
  country      year codes      n
  <chr>      <dbl> <chr>    <dbl>
1 Afghanistan 1980 new_sp_m014 NA
2 Afghanistan 1980 new_sp_m1524 NA
3 Afghanistan 1980 new_sp_m2534 NA
4 Afghanistan 1980 new_sp_m3544 NA
5 Afghanistan 1980 new_sp_m4554 NA
6 Afghanistan 1980 new_sp_m5564 NA
7 Afghanistan 1980 new_sp_m65 NA
8 Afghanistan 1980 new_sp_f014 NA
9 Afghanistan 1980 new_sp_f1524 NA
10 Afghanistan 1980 new_sp_f2534 NA
" # A tibble: 405,440 × 4"
```

separate() / unite()

```
1 separate(  
2   <DATA>,  
3   <VARIABLE>,  
4   into = c("<VARIABLE1>", "<VARIABLE2>")  
5 )  
6  
7 unite(<DATA>, <VARIABLES>)
```

Your Turn 4

Use the **cases** data below. Separate the **sex_age** column into **sex** and **age** columns.

```
1 cases <- tribble(  
2   ~id,      ~sex_age,  
3   "1",      "male_56",  
4   "2",      "female_77",  
5   "3",      "female_49"  
6 )  
7 separate(_____, _____, into = c("_____", "_____"))
```

Your Turn 4

```
1 cases <- tribble(  
2   ~id,      ~sex_age,  
3   "1",      "male_56",  
4   "2",      "female_77",  
5   "3",      "female_49"  
6 )  
7 separate(cases, sex_age, into = c("sex", "age"))
```

```
# A tibble: 3 × 3  
  id      sex      age  
<chr> <chr> <chr>  
1 1      male    56  
2 2      female  77  
3 3      female  49
```


Your Turn 4

```
1 cases <- tribble(  
2   ~id,      ~sex_age,  
3   "1",      "male_56",  
4   "2",      "female_77",  
5   "3",      "female_49"  
6 )  
7 separate(  
8   cases,  
9   sex_age,  
10  into = c("sex", "age"),  
11  convert = TRUE  
12 )
```

```
# A tibble: 3 × 3  
  id      sex      age  
  <chr> <chr>   <int>  
1 1      male     56  
2 2      female   77  
3 3      female   49
```

Your Turn 5: Challenge!

There are two CSV files in this folder containing SEER data in breast cancer incidence in white and black women. For both sets of data:

Import the data

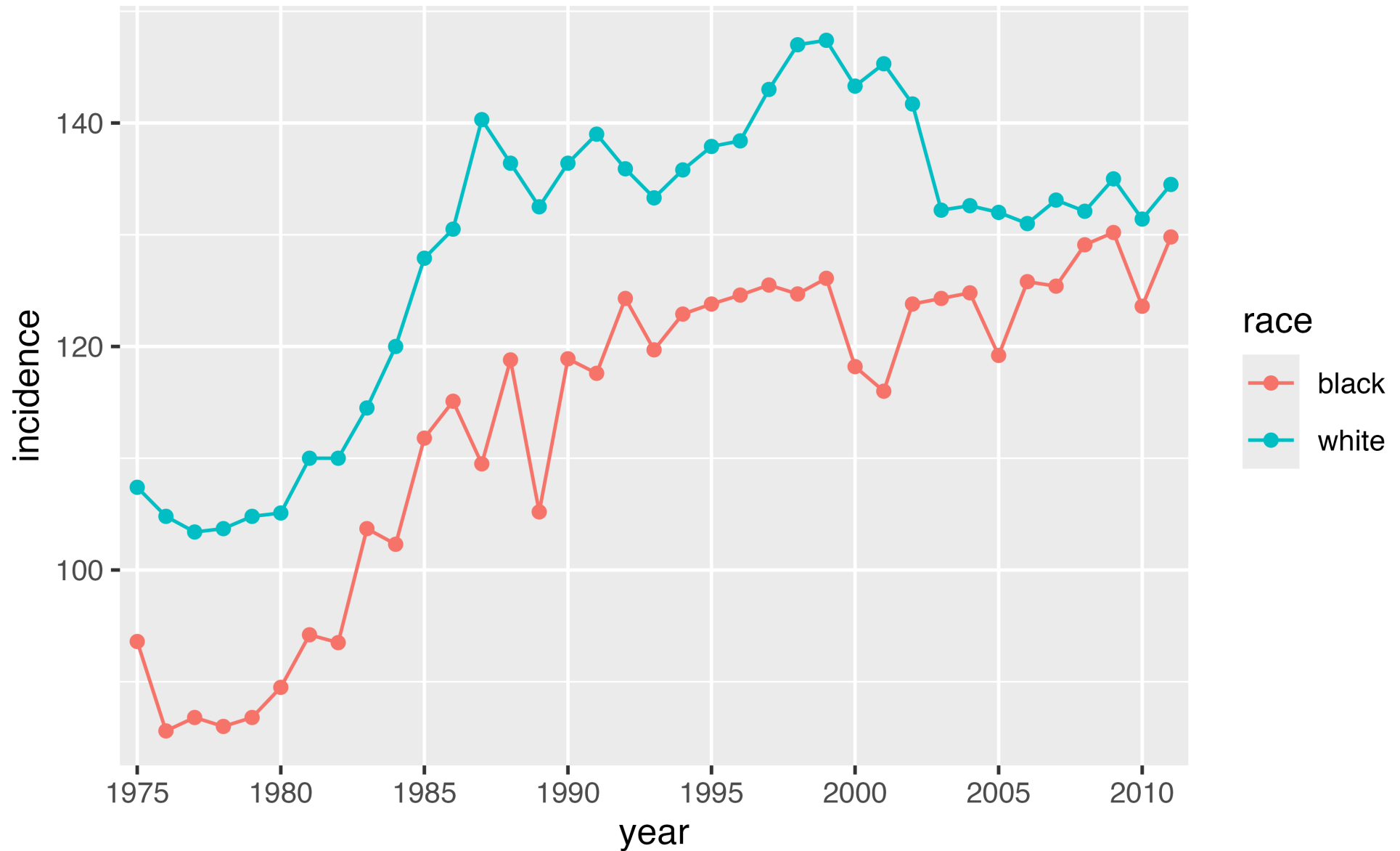
Pivot the columns into 2 new columns called `year` and `incidence`

Add a new variable called `race`. Remember that each data set corresponds to a single race.

Bind the data sets together using `bind_rows()` from the dplyr package. Either save it as a new object or pipe the result directly into the ggplot2 code.

Plot the data using the code below. Fill in the blanks to have `year` on the x-axis, `incidence` on the y-axis, and `race` as the color aesthetic.

Your Turn 5: No solution 🍆



Other neat tidyr tools

Uncounting frequency tables

```
1 lotr |>
2   pivot_longer(
3     names_to = "sex",
4     values_to = "count",
5     cols = c(female, male)
6   ) |>
7   uncount(count)
```

Other neat tidyr tools

```
# A tibble: 21,245 × 3
```

	film	race	sex
	<chr>	<chr>	<chr>
1	The Fellowship Of The Ring	Elf	female
2	The Fellowship Of The Ring	Elf	female
3	The Fellowship Of The Ring	Elf	female
4	The Fellowship Of The Ring	Elf	female
5	The Fellowship Of The Ring	Elf	female
6	The Fellowship Of The Ring	Elf	female
7	The Fellowship Of The Ring	Elf	female
8	The Fellowship Of The Ring	Elf	female
9	The Fellowship Of The Ring	Elf	female
10	The Fellowship Of The Ring	Elf	female
" : 34 335 more rows			

Other neat tidyr tools

Work with data frames

`crossing()` and `expand()`

`nest()` and `unnest()`

Other neat tidyr tools

Work with missing data

`complete()`

`drop_na()` and `replace_na()`

Resources

R for Data Science: A comprehensive but friendly introduction to the tidyverse. Free online.

Posit Recipes: Common code patterns in R (with some comparisons to SAS)